19 Given that $9^{-\frac{1}{2}} = 27^{\frac{1}{4}} \div 3^{x+1}$
find the exact value of x.

9^(-1/2) = (3^2)^(-1/2) = 3^-1.

27^(1/4) = (3^3)^(1/4) = 3^(3/4).

3^-1 = 3^(3/4) ÷ 3^(x+1) = 3^(-x-1/4).

-1 = -x - 1/4, so x = 3/4.

3 marks total
Convert to base 3
Use laws of indices
x = ¾

$x =$ .....

(Total for Question 19 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA